

DATA SPECIFICATION DOCUMENT

AI-Based Detection and Classification of Industrial Fires and Persistent Thermal Sources

Item	Details
Problem Statement ID	26162
Organization	National Technical Research Organisation (NTRO)
Document Version	1.0
Document Type	Data Specification

1. Purpose

This document defines the datasets required for the project, their sources, formats, fields, spatial and temporal properties, data quality requirements, and the structure of the integrated dataset.

It focuses only on data specifications. System architecture, product requirements, ML methodology, feature engineering, model selection, and deployment are covered in their respective documents.

2. Data Source Inventory

Data Source	Data Type	Format	Priority
NASA FIRMS	Thermal anomaly observations	CSV / JSON / SHP	Required
Historical NASA FIRMS	Historical thermal observations	CSV / JSON / SHP	Required
OpenStreetMap (OSM)	Industrial/geospatial features	PBF / GPKG / API	Required
ESA WorldCover	Land-cover raster	GeoTIFF	Recommended
Sentinel-2	Satellite imagery	Raster imagery	Optional / Advanced
Verified Labels	Classification labels	CSV / Parquet	Required for ML

3. NASA FIRMS Data Specification

3.1 Source

NASA Fire Information for Resource Management System (FIRMS).

3.2 Required Fields

Field	Data Type	Description
latitude	FLOAT	Latitude of detection
longitude	FLOAT	Longitude of detection
acquisition_date	DATE	Date of detection
acquisition_time	TIME	Time of detection
satellite	STRING	Satellite/platform
confidence	NUMERIC	Detection confidence
brightness	FLOAT	Brightness temperature

Field	Data Type	Description
frp	FLOAT	Fire Radiative Power

The exact fields may vary depending on the selected FIRMS product.

3.3 Metadata to Retain

```
product_name
product_version
satellite
download_date
start_date
end_date
geographic_extent
source_file
```

3.4 Spatial Representation

```
Geometry: Point
CRS: EPSG:4326
```

4. Historical FIRMS Data Specification

Historical FIRMS data will use the same core structure as the primary FIRMS dataset.

4.1 Required Information

```
latitude
longitude
acquisition_date
acquisition_time
satellite
confidence
brightness
frp
```

4.2 Additional Metadata

```
historical_period
source_file
download_date
product_version
```

4.3 Temporal Coverage

Initial development should use approximately 1–3 months of historical FIRMS data for the selected study region.

5. OpenStreetMap Data Specification

5.1 Source

OpenStreetMap (OSM), using an appropriate regional extract or targeted query.

5.2 Required Feature Categories

The project should extract relevant mapped industrial/geospatial objects, including industrial facilities, industrial areas, power plants, refineries, mines, quarries, storage facilities, and other relevant industrial infrastructure.

The exact OSM tags must be validated for the selected region before implementation.

5.3 Required Attributes

Attribute	Type	Description
osm_id	STRING	OSM object identifier

Attribute	Type	Description
feature_type	STRING	Type of facility/feature
name	STRING	Facility name, if available
geometry	GEOMETRY	Point, polygon, or other supported geometry
tags	JSON	Relevant OSM tags

5.4 Possible OSM Sources

```
Regional .osm.pbf
Regional .gpkg
Targeted OSM query
```

6. ESA WorldCover Data Specification

6.1 Source

ESA WorldCover.

6.2 Product Properties

Property	Specification
Resolution	10 m
Format	Cloud Optimized GeoTIFF
CRS	EPSG:4326
Available Product Years	2020 / 2021
Data Type	Raster
Coverage	Selected study region

6.3 Required Information

```
tile_id
product_year
land_cover_class
raster_path
crs
resolution
```

The land-cover class codes must be retained according to the selected WorldCover product's official classification scheme.

7. Sentinel-2 Data Specification

7.1 Source

Copernicus Sentinel-2 imagery.

7.2 Priority

Optional / Advanced. Sentinel-2 should initially be collected selectively for representative events rather than downloading large volumes of imagery.

7.3 Required Metadata

```
scene_id
acquisition_date
tile_id
cloud_cover
latitude
```

```
longitude
product_id
processing_level
```

7.4 Image Requirements

```
bands_used
spatial_resolution
image_format
scene_extent
cloud information
```

The exact bands and image-patch specifications will be finalized during implementation.

8. Verified Label Data Specification

8.1 Required Fields

Field	Type	Description
event_id	STRING	Associated thermal event
label	STRING	Verified class
verification_status	STRING	Verification state
verification_source	STRING	Source used for verification
verification_date	DATE	Date of verification
notes	TEXT	Optional verification notes

8.2 Candidate Classes

```
industrial_fire
persistent_industrial_source
wildfire
agricultural_burning
mining_related
unknown
```

The final label taxonomy will be finalized before creation of the training dataset.

9. Unified Event Data Schema

After source-specific processing, the system will maintain a common thermal-event record.

```
event_id
latitude
longitude
timestamp

frp
brightness
confidence
satellite

osm_facility_id
osm_feature_type

land_cover_class

sentinel2_scene_id
sentinel2_acquisition_date

label
```

10. Source-to-Event Relationships

Source	Relationship
FIRMS	Primary thermal observation
Historical FIRMS	Historical observations associated with the event/location
OSM	Nearby/relevant geographic features
WorldCover	Land-cover information for event location
Sentinel-2	Satellite scene associated with event location/date
Labels	Verified classification associated with event

11. Spatial Specification

11.1 Primary Coordinate Reference System

```
EPSG:4326 — WGS 84
```

11.2 Geometry Types

```
FIRMS:
Point

OSM:
Point / Line / Polygon

WorldCover:
Raster

Sentinel-2:
Raster
```

Spatial data must retain its original CRS metadata and be converted where necessary for processing.

12. Temporal Specification

All timestamps should be normalized to a consistent standard, preferably UTC.

```
date
time
timestamp
source
```

The original acquisition date/time should not be discarded during preprocessing.

13. Data Quality Requirements

13.1 FIRMS

Latitude must be between -90 and 90. Longitude must be between -180 and 180. Timestamp must be valid. FRP must contain valid numeric values. Confidence must conform to the selected FIRMS product. Duplicate observations should be identified.

13.2 OSM

Geometry must be valid. OSM identifiers must be retained. Feature types must be mapped consistently. Regional OSM tagging must be checked.

13.3 WorldCover

Correct tile must be used. CRS and resolution must be verified. Raster must be readable. Land-cover values must belong to the product's valid classes.

13.4 Sentinel-2

Scene must cover the required location. Acquisition date must be recorded. Cloud information must be retained. Invalid or corrupted scenes must be excluded.

14. Missing Data

Missing values must be explicitly represented as NULL. The system must distinguish between missing data and an actual zero value. No missing value should be replaced with an arbitrary value without documenting the transformation.

15. Duplicate Handling

Duplicate records should be detected using an appropriate combination of source, latitude, longitude, acquisition date, acquisition time, and satellite. Original raw records should be preserved even when duplicates are removed from processed datasets.

16. Data Versioning

Each dataset should maintain a source name, product name, product version, download date, processing version, geographic region, and temporal range.

```
Example:
FIRMS:
VIIRS__

OSM:
—

WorldCover:
WorldCover_

Sentinel-2:

Derived:
dataset_v1
```

17. Raw and Processed Data

Raw source data must be preserved separately from processed data. Raw files must not be modified directly.

```
data/
■
■■■ raw/
■ ■■■ firms/
■ ■■■ osm/
■ ■■■ worldcover/
■ ■■■ sentinel2/
■
■■■ processed/
■ ■■■ firms/
■ ■■■ osm/
■ ■■■ worldcover/
■ ■■■ sentinel2/
■
■■■ labels/
```

18. Initial Acquisition Parameters

Parameter	Value
Study region	TBD
FIRMS product	TBD
FIRMS satellite	TBD
FIRMS time period	TBD
OSM extract/query	TBD
WorldCover year	TBD
Sentinel-2 product	TBD
Sentinel-2 bands	TBD
Data version	v1.0

The initial development strategy should use one industrial region and approximately 1–3 months of FIRMS data, followed by matching OSM/WorldCover data and selective Sentinel-2 imagery.

19. Final Data Deliverables

1. Raw FIRMS dataset
2. Historical FIRMS dataset
3. OSM industrial/geospatial dataset
4. WorldCover raster dataset
5. Selected Sentinel-2 scenes
6. Verified label dataset
7. Unified event dataset
8. Dataset metadata/version records

The resulting unified dataset will be handed over to the ML/AI pipeline for feature engineering and model development.

20. Document Boundary

This document intentionally does not define system architecture, frontend/backend architecture, API design, database architecture, ML algorithms, feature-engineering formulas, classification models, training procedures, model evaluation strategy, prediction pipeline, GIS dashboard design, or alerting mechanisms. Those belong to the PRD, System Architecture, and ML/AI Design documents.

This keeps the Data Specification focused strictly on what data exists, where it comes from, what its structure is, and what quality it must satisfy.